Smart Waste Collection Using Ant Colony Optimization (ACO)

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***Abstract*—In large urban settings which are see to be growing by the day traditional collection systems which we see to be on fixed routes without looking at the present state of waste bins are put in place. This does in fact lead to many unneeded trips, high fuel use, and some bins over flowing. To solve these issues our project has put forth a smart waste collection system which we have designed through data driven decision making. In this system we use simulated sensor data to determine the fill levels of the bins out which we in turn we present as percent full which in the end we use to determine the priority of each bin. Based in that which we have determined bins are put into categories and from there an optimal collection route is designed which gives preference to the fully filled bins. We use a graph based model which displays how the bins relate to each other which in turn helps us determine the best collection routes. This we think makes the operation of the system easy to understand. As a whole what we have put forth reduces unnecessary travel, saves time and resources and we put forth a practical solution for smarter waste management in urban settings.**

I. Introduction

Waste management in today's urban environment is a large issue. That is in large part due to a growing population that is putting out more waste daily. Also in a great deal of our areas we see the same old waste collection routines used set in stone routes and times. That in turn means at times we see full bin collections that could have been put off a while yet. At the same time other bins are filling up and in some cases over flowing before the next collection round. Which in turn means we have an inefficient service that also is a waste of fuel, of staff time, and leaves us with a less than clean environment. What is needed is a better solution which is able to react to the real time state of the bins. In this project we put forth a basic smart waste collection system. In this project we developed a simple smart waste collection system. We did not follow fixed routes in the system which instead looks at how full each bin is and gives priority to almost full bins. Thus the collection process became more efficient and we avoided unnecessary trips. We did not use real sensors in our case, instead we created some sample (simulated) data to represent the bin conditions. We used this data for the system to determine the fill level of each bin. Also we had the system determine the order in which to collect the bins. Also we included a graph which shows how the bins are connected which in turn helps in better understanding of the route.

A. Scope

The main aim of this project is to design and show a simple smart waste collection system that can improve the way waste is usually managed. It combines basic ideas like sensor data handling, route optimization, and visualization into one system. The goal is not to build a full real-world system, but to give a clear idea of how such systems can work in practical situations.

In this project, real sensors are not used. Instead, we use simulated data to represent how full each waste bin is. This makes it easier to develop and test the system without needing actual hardware. The system then processes this data and converts it into useful information like fill percentage and priority level. Based on this, it decides which bins should be collected first and plans a better route. The system is designed on a small scale, with only a few bins and simple routes. It does not include advanced features like GPS tracking or large-scale deployment. Even though it is simple, it still helps in understanding how smart waste collection can work. Overall, the project mainly focuses on simulation, basic optimization, and visualization. Even with these limitations, it clearly shows how smart systems can help improve efficiency and reduce unnecessary effort in waste collection.

B. Objective

The main aim of this project is to develop a smart waste collection system that can improve the efficiency of waste management. Instead of following fixed schedules, the system uses data to make better decisions. This helps in reducing unnecessary trips and makes sure that bins are collected only when needed.

One of the important objectives of this project is to estimate how full each bin is using simulated sensor data. Based on this, the system assigns priority to each bin. Bins that are almost full are given more importance so that they can be cleared first. Another objective is to find a better route for waste collection so that the distance travelled and time taken can be reduced. The project also focuses on showing the working of the system in a simple and clear way. A basic interface is used to display bin details and the optimized route, so it becomes easier to understand how the system is working. In addition, a graph-based approach is used for route planning, which is a useful concept in many real-life applications.

Overall, the aim is to build a simple and useful system that shows how technology can improve traditional waste collection methods.

II. RELATED WORK

In earlier systems, waste collection was usually done using fixed routes. This method was not very efficient because sometimes bins were empty when collected, and in other cases, they were already full. Because of this, there was unnecessary travel, which increased both time and cost.

To improve this situation, researchers started using optimization techniques like Ant Colony Optimization (ACO). This method is based on how ants find the shortest path to food. In a similar way, ACO helps in finding better routes for waste collection, which reduces travel distance and saves time. Some systems also make use of sensors to check how full the bins are. This helps in identifying which bins need to be collected first. When sensor data is combined with optimization methods, the overall efficiency of the system becomes better. In some recent works, machine learning techniques are also used to predict waste levels in advance. But these methods can be a bit complex and may require more resources. In this project, a simpler approach is used by applying ACO for route optimization. The system is easy to implement and works well for small-scale setups, while still improving efficiency compared to traditional methods.

A. Smart waste management system

In recent years, smart waste management has become more important because cities are growing and need better services. In many places, waste is still collected based on fixed schedules, without checking the actual condition of the bins. Because of this, some bins are collected even when they are not full, while others overflow. This makes the system less efficient and also increases cost and environmental problems.

To improve this, some systems have been developed that use sensors to check how full the bins are. These sensors can send alerts when a bin reaches a certain level. This reduces manual work and helps in responding faster. However, most of these systems mainly focus on monitoring and giving alerts, and not much on improving the collection process itself. Also, in some cases, people still need to manually decide the collection routes, which reduces the overall efficiency. In this project, we try to improve on these ideas by not only checking bin levels but also deciding which bins should be collected first and finding a better route automatically. By combining both monitoring and decision-making, the system becomes more complete and useful.

B. Internet of(IOT) things in waste collection

The Internet of Things (IoT) has helped improve waste management systems by making them smarter and more responsive. In many IoT-based systems, sensors like ultrasonic sensors are used to measure how full a bin is. This is done by checking the distance between the waste and the top of the bin. The collected data is then sent to a central system for further processing. These systems make it possible to monitor bins in real time, which helps in identifying the ones that need immediate collection. But in real-world use, IoT systems also have some challenges. Setting up sensors can be costly, and maintaining them is not always easy. In addition, these systems depend on proper network connectivity, and any failure in sensors or communication can affect how well the system works. In this project, instead of using real sensors, we use simulated data to represent the condition of the bins. This makes the system easier to develop and test without spending much on hardware. It also allows us to try different situations easily. Even though it is a simulation, it still gives a good idea of how an actual IoT-based waste management system would work.

C. Route optimization techniques

Route optimization is an important part of improving waste collection systems. In different research works, many algorithms have been used to solve routing problems. Some common ones are Dijkstra's algorithm for finding the shortest path, Genetic Algorithms for optimization, and Ant Colony Optimization, which is based on the behavior of ants. These methods are useful, especially when there are many locations involved. However, these techniques can sometimes be a bit complex. They may require more calculations and careful tuning of parameters. For small projects or simple demonstrations, using such methods may not always be necessary. In many cases, simpler approaches can still give good results without making the system too complicated. In this project, a basic priority-based method is used to decide the order of bin collection. Bins that are more full are given higher priority, so they are collected first. Even though this method is simple, it still shows how route optimization works. It can also be improved later by using more advanced algorithms if needed.

D. Equations

The system uses a simple mathematical model to calculate the fill level of each waste bin. The fill percentage is deter-

mined based on the distance measured between the waste level and the top of the bin.

The formula used is:

$$Fill\ Percentage = \frac{(H-d)}{H} \times 100 \tag{1}$$

where H represents the maximum height of the bin and d represents the measured distance from the sensor to the waste level.

A lower distance value indicates that the bin is nearly full, while a higher distance value indicates that the bin is empty or partially filled.

Based on the calculated fill percentage, bins are categorized into different priority levels as follows:

$$Priority = \begin{cases} High, & \text{if } Fill > 80 \\ Medium, & \text{if } 50 < Fill \leq 80 \\ Low, & \text{if } Fill \leq 50 \end{cases} \tag{2}$$

These equations help in transforming raw sensor data into meaningful information, which is then used for route optimization and decision making in the system.

E. Graph based models

Graph-based models are commonly used when a system has many connected parts. In this type of model, nodes are used to represent things, and edges show how they are connected. This method is useful in problems related to transportation and routing. In waste management, bins can be treated as nodes, and the roads between them can be shown as edges with distances. This makes it easier to look at different possible routes and choose a better one. Many research works have used graph-based methods for solving routing problems because it gives a clear and structured way to represent the system. In this project, a graph model is used to show the waste collection network. It helps in finding better routes and also makes the system easier to understand visually. Instead of just using data, the graphical view gives a clearer idea of how everything is connected.

F. Visualization techniques

Visualization is very important when it comes to understanding how a system works. If we only look at raw data, it can be difficult to understand what is actually happening. Showing the data in a graphical form makes it much easier to interpret and analyze. There are many tools available to create such visualizations, like dashboards and different libraries. In this project, Streamlit is used to build a simple interface. It shows the fill levels of bins, the optimized route, and also a graphical view of the network. This makes the system easier to follow. Visualization also helps a lot during presentations. Instead of just explaining the system, we can show how it works step by step. It becomes easier for others to understand how the data is processed and how decisions are made. Overall, it makes the system more clear and user-friendly.

An excellent style manual for science writers is [7].

G. Limitation of existing system

Even though smart waste management systems have improved a lot, there are still some limitations in existing solutions. Many of them depend on hardware like sensors and communication devices, which can increase the cost and make the system more complicated. Maintaining these devices can also be difficult, especially when the system is used on a large scale.

Some systems only focus on checking how full the bins are, but they do not consider route optimization. Because of this, the overall efficiency is still limited. On the other hand, some methods use complex algorithms, which may not be suitable for small projects since they require more computation and setup.

Another issue is that many systems are not flexible enough to handle changes, like sudden increases in waste or unexpected situations. In this project, we try to keep things simple by using a combination of simulation, prioritization, and visualization. It is not a full real-world system, but it gives a clear idea of how such systems can be improved and extended in the future.

III. DATA INPUT PROCESSING

A.Data overview In this project, real sensor data is not used. Instead, we use simulated data to represent how full each waste bin is. This makes it easier to show how the system works without using actual hardware. The data represents the distance between the waste level and the top of the bin, which is similar to how an ultrasonic sensor works.

Each bin is given a distance value within a certain range, usually between 5 cm and 30 cm. If the distance is small, it means the bin is almost full. If the distance is larger, the bin is either empty or only partly filled. Using this kind of data helps the system behave almost like a real setup.

The bins are named as Bin1, Bin2, and so on, and they are connected using a fixed network structure. This structure is later used to decide the route for collection. Even though the data is not real, it still shows how waste levels can change and helps in testing different situations.

A. Data handling and selection

In this project, we are not using any machine learning model, so there is no need to divide the data into training and testing sets. Instead, the system works directly with the input data, which can be real-time or simulated. Every time the system runs, new values are generated to represent changes in the bin conditions. The system checks how full each bin is and then arranges them based on priority. Bins that are more full are given higher priority and are selected first. After selecting these bins, the system decides the order in which they should be collected and generates a better route. This way of handling data makes the system more flexible, since it can easily adjust to changing conditions without depending on past data. It also keeps the process simple, as decisions are made directly using the current data instead of going through complex training steps.

IV. METHODOLOGY

A. System model

The proposed system is designed as a simple smart waste management model that combines data input, processing, and visualization. It mainly has four parts: data input, processing, decision-making, and visualization. At the beginning, bin data is generated using simulated sensor values. These values represent the distance between the waste level and the top of the bin. The system then processes this data and calculates how full each bin is in terms of percentage. Based on these values, the bins are grouped into different priority levels. Bins that are more full are given higher priority compared to others. To represent the system clearly, a graph model is used where bins are shown as nodes and the paths between them are shown as edges. This helps in understanding how the bins are connected and also supports route planning. Finally, the output is shown through a simple interface where users can see the bin status, their priority, and the optimized route. The system follows a step-by-step process from data input to final display, which makes it easy to understand and implement.

B. Algorithm and working

The system follows a simple and efficient approach to determine the optimized waste collection route. The steps involved are as follows:

1) Generate bin data using simulated values
2) Calculate fill percentage for each bin
3) Assign priority based on fill level
4) Sort bins in descending order of priority
5) Create a route starting from the depot and ending at the waste center
6) Visualize the system using a graph representation

The fill percentage is calculated using a basic formula that compares the measured distance with the maximum bin height. Bins with higher fill percentages are considered more critical and are selected first.

A graph is used to represent the system, where each node corresponds to a bin, and edges represent the connections between them. The optimized route is highlighted on the graph, making it easier to understand the collection path. This approach is simple, efficient, and suitable for small-scale systems. It can also be extended to include more advanced optimization techniques in future work.

C. System flow diagram

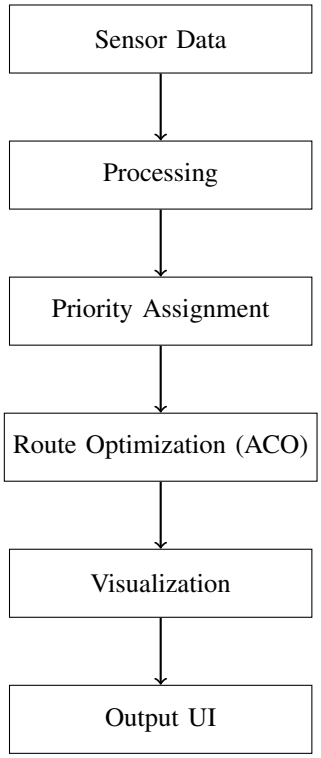


Fig. 1. System flow diagram of smart waste collection using ACO

V. EXPERIMENTAL RESULTS

The developed system was tested using simulated bin data to check how well it works. Every time the system runs, new values are generated for the bins to represent changing conditions. Based on these values, the system calculates how full each bin is and assigns priority to them.

The system is able to generate a route starting from the depot and ending at the waste collection center. Bins that are more full are given higher priority, so they are collected first. The final result is shown through a simple interface, which makes it easier to understand how the system is working.

A. Graphical representation

A graph-based view is used in this project to show how the waste collection system works. In this graph, each bin is shown as a node, and the paths between the bins are shown as edges. These edges also have distance values, which represent how far one bin is from another. Different colors are used to show how full each bin is. Green means the bin has a low fill level, orange shows a medium level, and red indicates that the bin is almost full. The final route selected by the system is also highlighted, so it is easy to see which path is chosen. This kind of visual representation makes it easier to understand how the system gives priority to bins and decides the route. It helps in clearly seeing the working of the system instead of just looking at raw data.

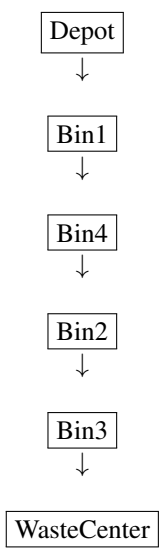
B. Performance analysis

The system is able to generate routes based on the input data, which keeps changing each time it runs. Since the data is generated dynamically, the system can adjust to different conditions and produce different routes for different situations. This makes the system flexible and helps in improving the overall collection process. When compared to traditional systems that use fixed routes, this approach reduces unnecessary travel and works more efficiently. By giving priority to bins that are more full, the system makes sure that important bins are collected first, which also helps in avoiding overflow. The system also includes a simple and clear visual output. This makes it easier to understand how everything is working and is especially useful during demonstrations. Sample Output Table

C. Sample Output

Bin	Distance (cm)	Fill (%)
Bin1	6	80%
Bin2	15	50%
Bin3	25	16%
Bin4	10	66%

TABLE I
SAMPLE BIN DATA AND CALCULATED FILL LEVELS



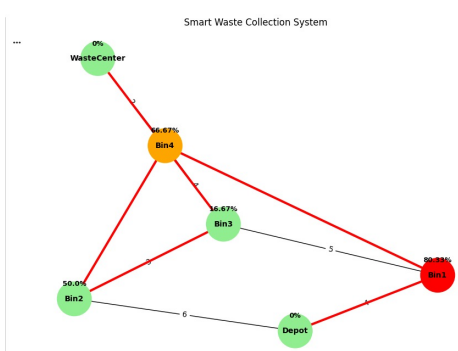


Fig. 2. Graph representation of waste collection network

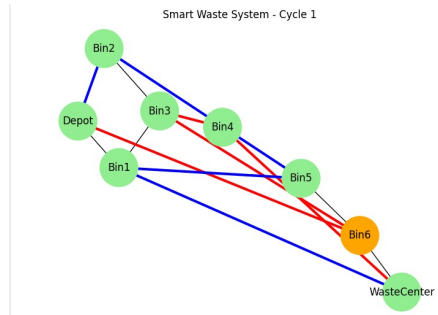


Fig. 3. ACO-based optimal path for single vehicle routing

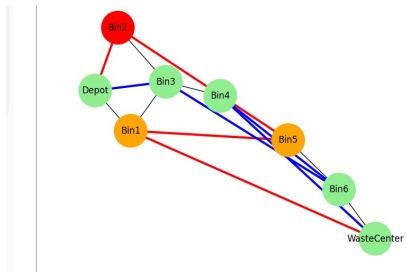


Fig. 4. ACO-based multi-vehicle route optimization

VI. DISCUSSION

A. Key findings

The proposed smart waste collection system shows how simple data-based methods can improve the way waste is managed. The system takes simulated sensor data, converts it into fill levels, and then uses that to decide which bins should be collected first. Because of this, bins that are more full are given priority, which helps in reducing overflow. The graph-based model used in the system gives a clear idea of how the bins are connected. It makes it easier to understand the network and also helps in selecting a better route. Since the input data changes each time, the system is able to adjust and generate different routes, which shows that it can handle changing conditions. One important point is that even a simple priority-based approach can give good results without using complex algorithms. Also, the use of a basic interface makes the system easier to understand and use, especially for demonstration or learning purposes..

B. Limitation and future work

Although the system performs effectively, there are certain limitations that need to be considered:

1) **Use of Simulated Data:** The system relies on generated values instead of real sensor data. While this is useful for demonstration, real-world implementation would require physical sensors.
2) **Limited Scale:** The current system is designed for a small number of bins. Scaling it to larger environments may require more advanced optimization techniques.

3) **No Real-Time Tracking:** The system does not include GPS-based tracking or live monitoring of collection vehicles.
4) **Simplified Routing Approach:** The routing method used is based on priority sorting rather than advanced algorithms like Ant Colony Optimization or Dijkstra's algorithm.

To improve the system, the following enhancements can be considered:

1) Integration with real IoT sensors for live data collection
2) Use of advanced algorithms for better route optimization
3) Development of a mobile application for real-time monitoring
4) Expansion of the system for large-scale smart city deployment

VII. CONCLUSION

The proposed smart waste collection system gives a simple idea of how waste management can be improved using data. In this project, simulated sensor data is used to estimate how full each bin is, and based on that, the bins are given priority. This helps in making sure that bins which need attention are cleared first, so overflow can be avoided.The use of a graph model makes it easier to understand how the bins are connected and helps in generating a better route. The system also has a simple interface where the bin status and route can be seen clearly. This makes the system easier to use and also helps in understanding how it works.One of the main advantages of this project is that it is simple. It does not use complex algorithms or large datasets, but still shows how data can be used to make better decisions in waste collection. It also gives an idea of how IoT concepts and optimization can be applied in real situations. Overall, this project gives a basic foundation for building smarter waste management systems. In the future, it can be improved by using real sensors and more advanced routing methods, so that it can be used in real-world smart city applications.

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